

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 7: Ratio and Proportion

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Properties of proportion

1. Use the properties of proportion to solve $\frac{6x^2 + 3x - 5}{3x - 5} = \frac{9x^2 + 2x + 5}{2x + 5}$, where $x \neq 0$.
ICSE 2025, 3 marks
2. Given $\frac{(a+b)^3}{(a-b)^3} = \frac{64}{27}$, find (a) $\frac{a+b}{a-b}$ (b) the ratio $a : b$, using the properties of proportion.
ICSE 2024, 4 marks
3. Use componendo and dividendo to solve $\frac{\sqrt{2x+2} + \sqrt{2x-1}}{\sqrt{2x+2} - \sqrt{2x-1}} = 3$. ICSE 2023, 3 marks
4. Find $x : y$, given $\frac{x^2 + 2x}{2x + 4} = \frac{y^2 + 3y}{3y + 9}$. ICSE 2020, 3 marks
5. If $x = \frac{\sqrt{2a+1} + \sqrt{2a-1}}{\sqrt{2a+1} - \sqrt{2a-1}}$, prove that $x^2 - 4ax + 1 = 0$. ICSE 2020, 4 marks
6. Use the properties of proportion to solve $\frac{\sqrt{5x} + \sqrt{2x-6}}{\sqrt{5x} - \sqrt{2x-6}} = 4$. ICSE 2019, 3 marks
7. Use the properties of proportion to solve $\frac{2x + \sqrt{4x^2 - 1}}{2x - \sqrt{4x^2 - 1}} = 4$, given that x is positive.
ICSE 2018, 3 marks
8. Given $\frac{7m + 2n}{7m - 2n} = \frac{5}{3}$, use the properties of proportion to find (a) $m : n$ (b) $\frac{m^2 + n^2}{m^2 - n^2}$.
ICSE 2017, 3 marks

2 Proportion

9. A train moves at a uniform speed. It covers 60 m in 2 s, 90 m in x s and y m in 5 s. Find x and y . (a) $x = 4, y = 150$ (b) $x = 3, y = 100$ (c) $x = 4, y = 100$ (d) $x = 3, y = 150$ ICSE 2024, 1 mark
10. What number must be added to each of 4, 6, 8 and 11 so that the four numbers are in proportion? ICSE 2023, 3 marks
11. The numbers $K + 3, K + 2, 3K - 7$ and $2K - 3$ are in proportion. Find K . ICSE 2019, 3 marks

3 Continued proportion

12. Find the mean proportional between 4 and 9. (a) 4 (b) 6 (c) 9 (d) 36 ICSE 2023, 1 mark

13. b is the mean proportional between a and c . Show that $\frac{a^4 + a^2b^2 + b^4}{b^4 + b^2c^2 + c^4} = \frac{a^2}{c^2}$. **ICSE 2017, 3 marks**

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